



Tezpur University

Theory and Implementation of Quantum Key Distribution:

A Comparative Study of BB84 and LM05

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Preface

Purpose of This Report

This report exists to answer a single question thoroughly: how do two different quantum key distribution protocols — BB84 and LM05 — actually behave, not just in theory, but when built as real circuits, run on simulators, and finally executed on genuine quantum hardware? Many treatments of quantum key distribution (QKD) stop at the theoretical level, presenting security proofs and error-rate formulas without ever showing what happens when those ideas are translated into working code and tested against the imperfections of the real world. This report takes the opposite approach: every claim made here is backed by an actual implementation, an actual simulation, or an actual measurement taken from real IBM quantum hardware.

The two protocols under study, BB84 and LM05, were chosen because they represent two genuinely different philosophies for achieving the same goal — a secret key shared between two parties, Alice and Bob, that no eavesdropper can learn without being detected. BB84, introduced by Bennett and Brassard in 1984, is a one-way protocol: a qubit travels once from Alice to Bob, and security emerges from the impossibility of measuring an unknown quantum state without a chance of disturbing it. LM05, introduced by Lucamarini and Mancini in 2005, takes a two-way approach: a qubit travels from Bob to Alice and back again, allowing Alice to imprint a message directly onto the returning qubit. These are not merely two variations of the same idea — they rest on structurally different mechanisms, and this report is, in large part, an investigation into how that structural difference manifests as a measurable difference in performance.

How to Read This Report

This report is written for a reader who already has some exposure to quantum computing — someone comfortable with the idea of a qubit, familiar with basic linear algebra, and aware that quantum mechanics is probabilistic in nature — but who is not assumed to already know quantum key distribution in depth. Every concept specific to QKD is built up from first principles. Nothing is assumed to be “obvious” simply because it is standard in the literature; if a step in an argument or a piece of mathematics is used, it is derived or explained at the point where it is first needed.

The report is organized into four parts. Part I lays the foundations: the physical and mathematical building blocks — qubits, bases, measurement, the no-cloning theorem — that

both protocols depend on. Part II develops the theory of BB84 and LM05 individually, in full mathematical detail, before comparing them directly. Part III is where theory becomes practice: circuit design, and then three levels of implementation — an ideal noise-free simulator, a realistic noisy simulator, and finally real IBM quantum hardware. Part IV synthesizes everything: a combined analysis across all three implementation levels, an honest accounting of this project’s limitations, and a final conclusion.

Throughout, mathematics is presented in full — state vectors, gate matrices, and probability calculations are written out explicitly rather than merely referenced, in keeping with the aim of this report: not simply to state what happens, but to show, step by step, why it happens.

Notation Conventions

Quantum states are written in Dirac (bra-ket) notation: $|0\rangle$ and $|1\rangle$ denote the two states of the computational (Z) basis, and $|+\rangle = \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle)$, $|-\rangle = \frac{1}{\sqrt{2}}(|0\rangle - |1\rangle)$ denote the two states of the diagonal (X) basis. Gate matrices are denoted by their standard symbols: X , H , Y , I for the Pauli-X, Hadamard, Pauli-Y, and Identity operators respectively. The quantum bit error rate is denoted QBER, and the secure key rate is denoted R_{SKR} . Throughout Part III, all numerical results quoted are taken directly from the author’s own implementation and are not illustrative or hypothetical.

Part I

Foundations

Chapter 1

The Problem of Secure Key Distribution

1.1 Why Classical Cryptography Needs a Shared Secret

Suppose two people, Alice and Bob, want to exchange a private message over a communication channel that anyone might be listening to. The most reliable way to guarantee privacy, known since long before computers existed, is the *one-time pad*: Alice encrypts her message by combining it, bit by bit, with a random string of bits — the *key* — that only she and Bob possess. Bob decrypts by applying the same operation in reverse. If the key is truly random, used only once, and known to no one but Alice and Bob, this scheme is mathematically unbreakable: an eavesdropper who intercepts the encrypted message learns nothing whatsoever about its content, no matter how much computational power they have.

The catch is immediate and severe: Alice and Bob must already share this secret key before they can communicate securely. But how did they get the key to each other in the first place, if not over some communication channel that could itself have been intercepted? This is not a minor technical inconvenience — it is the central, unavoidable problem that all of cryptography must eventually confront. Modern classical cryptography largely sidesteps this problem rather than solving it, using mathematical constructions (such as the difficulty of factoring large numbers, in RSA encryption) that allow two parties to agree on a shared secret without ever transmitting it directly. This works, but it rests on an assumption: that certain mathematical problems are hard to solve. If that assumption ever turns out to be false — for instance, if someone builds a sufficiently powerful quantum computer capable of factoring large numbers efficiently, using Shor's algorithm — the entire security guarantee collapses.

1.2 The Key Distribution Problem, Precisely Stated

The problem, stated precisely, is this: Alice and Bob are physically separated and can only communicate over channels that a third party, conventionally named Eve, may be able to

observe or even tamper with. They need to end up holding an identical string of random bits — a key — such that Eve, even if she recorded every single bit transmitted over the channel, cannot determine the key (or can determine, at most, a vanishingly small and quantifiable amount of information about it). Whatever protocol Alice and Bob use to achieve this must also let them detect if Eve was listening, so that a compromised key is never mistaken for a secure one.

Classical physics offers no way to guarantee this. A classical bit, once observed, can be recorded and copied perfectly without leaving any trace — there is no physical mechanism by which observing a classical signal must disturb it. If Eve intercepts a classical transmission, copies it exactly, and forwards an identical copy onward, Alice and Bob have no way of detecting that anything happened. This is the fundamental limitation that classical key distribution cannot overcome on its own, and it is precisely the limitation that quantum mechanics is able to break.

1.3 Why Quantum Mechanics Offers a Solution

Quantum mechanics introduces two properties that have no classical analogue, and together they make secure key distribution possible in a way that does not rely on any unproven computational hardness assumption.

The first property is that **measurement generally disturbs an unknown quantum state**. Unlike a classical bit, a quantum bit (qubit) can be prepared in a state that is not aligned with the basis in which it will eventually be measured. If Eve does not know which basis Alice used to prepare a qubit, and Eve is forced to guess a basis in order to extract any information from it, her measurement — when performed in the wrong basis — will on average disturb the qubit's state. This disturbance is not a matter of poor experimental technique; it is a direct consequence of the mathematics of quantum measurement (the Born rule, introduced in the next chapter), and it means that eavesdropping leaves a statistical trace.

The second property is the **no-cloning theorem**: it is physically impossible to create an identical, independent copy of an arbitrary unknown quantum state. This closes off the strategy that defeats classical key distribution — Eve cannot simply copy the qubit, keep one copy for herself, and forward an identical copy onward undetected, the way she could with a classical bit. Whatever she does to extract information from the qubit, she is stuck interacting with the original, and that interaction is what creates the detectable disturbance described above.

Taken together, these two properties mean that a protocol for exchanging quantum states between Alice and Bob can, at least in principle, allow them to detect eavesdropping through the introduction of measurable, unavoidable errors — bounding, with the confidence of physical law rather than computational hardness, how much information any eavesdropper could possibly have gained. This is the foundational idea behind quantum key distribution (QKD), and the remainder of Part I develops the precise physical and mathematical tools needed to turn this idea into an actual protocol. Chapters 2 through 4 then present the tools and the two protocols — BB84 and LM05 — that put those tools to work in two structurally different ways.

Chapter 2

The Quantum Building Blocks

2.1 The Qubit and Its State-Vector Representation

A classical bit can be in exactly one of two states, 0 or 1, and nothing in between. A qubit is fundamentally different: it is described by a state vector in a two-dimensional complex vector space, and this vector can point in a continuum of directions, not just two. The two special reference directions in this space are written $|0\rangle$ and $|1\rangle$, and represented as column vectors:

$$|0\rangle = \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \quad |1\rangle = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$$

A general qubit state $|\psi\rangle$ is written as a linear combination — a *superposition* — of these two:

$$|\psi\rangle = \alpha |0\rangle + \beta |1\rangle = \begin{pmatrix} \alpha \\ \beta \end{pmatrix}$$

where α and β are complex numbers called *amplitudes*, subject to a single constraint that keeps the state physically meaningful:

$$|\alpha|^2 + |\beta|^2 = 1$$

This constraint is called *normalization*. It exists because, as will be made precise in Section 2.4, $|\alpha|^2$ and $|\beta|^2$ are interpreted as probabilities, and probabilities describing all possible outcomes of an experiment must sum to exactly 1. It is worth pausing on how strange this already is compared to classical intuition: a qubit does not simply store "0 or 1 with some probability" as a classical random bit would; it stores a genuine vector, with amplitudes that can interfere constructively or destructively with each other, a phenomenon with no classical counterpart at all. This report will not need to make deep use of quantum interference, but it is important to understand that the state vector $|\psi\rangle$ is not merely a bookkeeping device for probabilities — it is the complete physical description of the qubit, and probabilities are only extracted from it at the final step, when a measurement is performed.

2.2 The Z Basis and the X Basis

A *basis* is a choice of two reference directions against which a qubit's state can be expressed and, crucially, measured. The vectors $|0\rangle$ and $|1\rangle$ introduced above form one such basis, conventionally called the *computational basis* or the *Z basis* (the name comes from the fact that these two vectors are the eigenstates of the Pauli-Z operator, which will not be needed explicitly in this report, but the naming convention is worth knowing since it appears throughout the QKD literature).

There is a second, equally valid basis, called the *diagonal basis* or the *X basis*, built from equal superpositions of $|0\rangle$ and $|1\rangle$:

$$|+\rangle = \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle) = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 1 \end{pmatrix}, \quad |-\rangle = \frac{1}{\sqrt{2}}(|0\rangle - |1\rangle) = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ -1 \end{pmatrix}$$

It is worth verifying that $|+\rangle$ and $|-\rangle$ genuinely form a valid basis in their own right: they are normalized ($|\frac{1}{\sqrt{2}}|^2 + |\frac{1}{\sqrt{2}}|^2 = \frac{1}{2} + \frac{1}{2} = 1$), and they are orthogonal to each other, meaning $\langle + | - \rangle = 0$. Any qubit state can be equally well written as a combination of $|0\rangle$ and $|1\rangle$, or as a combination of $|+\rangle$ and $|-\rangle$ — both are complete, valid descriptions of the same underlying vector, viewed from two different reference frames, in the same way a point on a two-dimensional plane can be described using either a standard (x, y) coordinate system or one rotated by 45 degrees.

This last point — that Z and X are two different, equally valid *descriptions* of the same space — is the single most important geometric fact underlying both BB84 and LM05. The state $|0\rangle$, when described in terms of the X basis, is not purely $|+\rangle$ or purely $|-\rangle$, but an equal mixture of both: substituting the definitions above, $|0\rangle = \frac{1}{\sqrt{2}}(|+\rangle + |-\rangle)$. This is the algebraic origin of the property described qualitatively in Chapter 1: a state prepared in one basis, when examined through the lens of the other basis, appears maximally uncertain.

2.3 Superposition and Measurement

Having established that a qubit's state is a vector that can point in a continuum of directions, the next question is what happens when that qubit is actually measured — that is, when someone attempts to extract classical information from it. This is where quantum mechanics departs most sharply from ordinary intuition.

Measurement is always performed *with respect to a chosen basis*. If a physicist chooses to measure a qubit in the Z basis, the measurement apparatus is only capable of returning one of two outcomes, corresponding to $|0\rangle$ or $|1\rangle$ — it cannot return "some value in between," no matter how the qubit's actual state vector was oriented before the measurement. If the qubit's state was already exactly $|0\rangle$ or exactly $|1\rangle$, the outcome is certain and matches that state. But if the qubit's state was, for example, $|+\rangle$ — an equal superposition of $|0\rangle$ and $|1\rangle$ — the measurement forces the qubit to "choose" one of the two basis outcomes, and the choice is fundamentally probabilistic, governed by the precise rule given in the next section.

This forced choice is often called the *collapse* of the state vector: before measurement, the qubit genuinely exists as a superposition; after measurement, it has been projected onto one

of the two possible outcomes of the chosen basis, and any information about the amplitudes that existed before the measurement is now irrecoverably lost from that single qubit. This is why, as previewed in Chapter 1, measuring in the "wrong" basis is destructive in a very literal sense — it does not merely fail to reveal the original information, it actively destroys the very superposition that carried that information.

2.4 The Born Rule

The precise mathematical rule governing measurement outcomes is called the *Born rule*, named after physicist Max Born. For a qubit in state $|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$, measured in the Z basis, the Born rule states that the probability of obtaining outcome 0 is $|\alpha|^2$, and the probability of obtaining outcome 1 is $|\beta|^2$:

$$P(\text{outcome} = 0) = |\langle 0|\psi\rangle|^2 = |\alpha|^2, \quad P(\text{outcome} = 1) = |\langle 1|\psi\rangle|^2 = |\beta|^2$$

Here $\langle 0|\psi\rangle$ denotes the inner product between the bra vector $\langle 0|$ (the conjugate transpose of $|0\rangle$) and the state $|\psi\rangle$ — mechanically, this inner product simply extracts the amplitude of the $|0\rangle$ component of $|\psi\rangle$. This is precisely why the normalization condition $|\alpha|^2 + |\beta|^2 = 1$ from Section 2.1 was necessary: it guarantees that the two measurement probabilities always sum to exactly 1, as any valid set of probabilities must.

Two extreme cases are worth making explicit, since they recur constantly throughout this report. If $|\psi\rangle = |0\rangle$ exactly (so $\alpha = 1, \beta = 0$), then $P(\text{outcome} = 0) = |1|^2 = 1$ and $P(\text{outcome} = 1) = |0|^2 = 0$ — the outcome is certain. This is the case of a *matched basis*: the state was prepared in the same basis it is being measured in, and the Born rule reduces to a deterministic prediction, with zero probabilistic spread. If instead $|\psi\rangle = |+\rangle = \frac{1}{\sqrt{2}}|0\rangle + \frac{1}{\sqrt{2}}|1\rangle$ is measured in the Z basis, then $P(\text{outcome} = 0) = \left|\frac{1}{\sqrt{2}}\right|^2 = \frac{1}{2}$ and likewise $P(\text{outcome} = 1) = \frac{1}{2}$ — a completely random, maximally uncertain outcome. This is the case of a *mismatched basis*, and it is the direct mathematical source of the 50% random-guessing behavior that underlies eavesdropping detection in both BB84 and LM05, as will be derived formally in Chapters 3 and 4.

2.5 The No-Cloning Theorem

The final foundational tool needed before the protocols themselves can be developed is the no-cloning theorem, already introduced qualitatively in Chapter 1. It is worth presenting its proof here in full, since the proof itself is short, elegant, and reveals precisely why cloning fails — not merely that it fails.

Suppose, for contradiction, that there existed a *universal cloning machine*: a unitary operation U acting on two qubits — the qubit to be copied, and a second "blank" qubit initialized to some fixed state $|0\rangle$ — such that for *any* input state $|\psi\rangle$, the machine produces two identical copies:

$$U(|\psi\rangle \otimes |0\rangle) = |\psi\rangle \otimes |\psi\rangle$$

Suppose this machine works correctly for two particular, distinct states, $|\psi\rangle$ and $|\phi\rangle$:

$$U(|\psi\rangle \otimes |0\rangle) = |\psi\rangle \otimes |\psi\rangle, \quad U(|\phi\rangle \otimes |0\rangle) = |\phi\rangle \otimes |\phi\rangle$$

Now consider what the machine must do when given, as input, an equal superposition of these two states, $|\chi\rangle = \frac{1}{\sqrt{2}}(|\psi\rangle + |\phi\rangle)$. Because U is a unitary operator, and all unitary operators are *linear*, the machine's action on this superposition input is forced to be the corresponding superposition of its two known outputs:

$$U(|\chi\rangle \otimes |0\rangle) = \frac{1}{\sqrt{2}} \left(U(|\psi\rangle \otimes |0\rangle) + U(|\phi\rangle \otimes |0\rangle) \right) = \frac{1}{\sqrt{2}} (|\psi\rangle \otimes |\psi\rangle + |\phi\rangle \otimes |\phi\rangle)$$

But if the machine were genuinely cloning $|\chi\rangle$, correct cloning would instead require the output to be $|\chi\rangle \otimes |\chi\rangle$. Expanding this directly:

$$|\chi\rangle \otimes |\chi\rangle = \frac{1}{2} (|\psi\rangle \otimes |\psi\rangle + |\psi\rangle \otimes |\phi\rangle + |\phi\rangle \otimes |\psi\rangle + |\phi\rangle \otimes |\phi\rangle)$$

This expression contains four cross-terms, while the linearity-forced result above contains only two. These two expressions are not equal in general (they are only equal in the trivial case $|\psi\rangle = |\phi\rangle$) — a direct contradiction. Therefore, no unitary operation can correctly clone two distinct, non-orthogonal, unknown quantum states simultaneously — and since a truly *universal* cloning machine would need to work for arbitrary, unknown input states (which could be any pair of distinct states), no universal quantum cloning machine can exist at all.

This proof is the rigorous foundation for the qualitative claim made in Chapter 1: an eavesdropper, faced with an unknown qubit state prepared in one of two possible non-orthogonal bases, cannot silently duplicate it, keep a copy, and forward an undisturbed original onward. She is mathematically forced to interact with — and, per the Born rule developed above, potentially disturb — the only copy that exists. With the qubit, the two bases, the Born rule, and the no-cloning theorem now all established rigorously, the tools needed to construct an actual quantum key distribution protocol are complete. Chapter 3 puts them to work in the construction of BB84.

Part II

The Two Protocols, In Theory

Chapter 3

BB84 — Bennett and Brassard's Protocol

3.1 Historical Context

BB84 was introduced by Charles Bennett and Gilles Brassard in 1984, at a conference in Bangalore, India, in a paper titled "Quantum Cryptography: Public Key Distribution and Coin Tossing" [1]. It was the first protocol to demonstrate, concretely, that the quantum-mechanical tools developed in Chapter 2 — superposition, the Born rule, and the no-cloning theorem — could be assembled into a working procedure for secure key distribution, rather than remaining abstract curiosities of quantum theory. The name of the protocol is simply a contraction of the authors' initials and the year of publication, a naming convention that recurs throughout the QKD literature (as will be seen again with LM05 in Chapter 4).

It is worth noting, before proceeding, that BB84's original 1984 paper introduced the protocol's basic structure and the qualitative argument for its security, but not a fully rigorous mathematical security proof against arbitrary quantum attacks — that rigorous treatment came more than a decade later, from other authors, and is discussed separately in Section 3.7. This distinction matters for an accurate understanding of the protocol's history: the elegant idea and its airtight mathematical justification were, in this case, separated by many years of subsequent research.

3.2 State Preparation and Basis Encoding

BB84 involves two parties, Alice and Bob, connected by two channels: a quantum channel, over which qubits are physically transmitted, and a classical channel, over which ordinary messages can be exchanged (and which is assumed to be public — readable by an eavesdropper Eve — but authenticated, meaning Eve cannot impersonate Alice or Bob on it).

For each round of the protocol, Alice performs two independent random choices: a classical bit $a \in \{0, 1\}$, and a basis choice $\theta_A \in \{Z, X\}$, each drawn independently and uniformly at random. She then prepares a qubit according to the following rule, using the gate matrices introduced in Chapter 2:

$$|\psi\rangle_{\text{encoded}} = H^{[\theta_A=X]} X^{[a=1]} |0\rangle$$

where the bracket notation $G^{[\text{condition}]}$ denotes "apply gate G if the condition holds, otherwise apply the identity." Concretely, this produces one of the four states from Chapter 2: $a = 0, \theta_A = Z \rightarrow |0\rangle$; $a = 1, \theta_A = Z \rightarrow |1\rangle$; $a = 0, \theta_A = X \rightarrow |+\rangle$; $a = 1, \theta_A = X \rightarrow |-\rangle$. Alice transmits this qubit to Bob over the quantum channel.

Bob, upon receiving the qubit, independently chooses his own random basis $\theta_B \in \{Z, X\}$ and measures the qubit in that basis, obtaining a classical outcome $b \in \{0, 1\}$ via the Born rule (Section 2.4). Because Bob's basis choice is made independently of Alice's, and without any knowledge of it, there are four equally likely combinations of (θ_A, θ_B) per round, exactly as discussed in the general basis-mismatch analysis of Chapter 2.

3.3 Basis Reconciliation (Sifting)

After all N rounds of quantum transmission have taken place, Alice and Bob perform a purely classical step over the public channel: they announce, for each round, only their *basis* choice — never the bit value. This is safe to do publicly, since knowledge of the basis alone (without the bit) reveals nothing about the transmitted bit itself.

Both parties then discard every round in which $\theta_A \neq \theta_B$. As established in Section 2.4, a mismatched-basis measurement yields a completely random outcome with $P(\text{outcome} = 0) = P(\text{outcome} = 1) = \frac{1}{2}$, carrying zero information about Alice's original bit — so nothing of value is lost by discarding these rounds. Since θ_A and θ_B are independent uniform binary random variables, $P(\theta_A = \theta_B) = \frac{1}{2}$, and so the number of surviving (*sifted*) bits K out of N raw transmissions follows a binomial distribution:

$$K \sim \text{Binomial}(N, p = 0.5), \quad \mathbb{E}[K] = \frac{N}{2}$$

The remaining K bits, held identically (in the absence of noise or eavesdropping) by both Alice and Bob, form the *sifted key* — the raw material from which the final secret key will eventually be distilled.

3.4 Quantum Bit Error Rate (QBER)

Even among the sifted (basis-matched) bits, Alice's and Bob's values may disagree, either due to physical noise in the channel and hardware, or due to disturbance introduced by an eavesdropper. The fraction of such disagreements is the Quantum Bit Error Rate:

$$\text{QBER} = \frac{\text{number of mismatched bits in a compared sample}}{\text{total number of bits compared}}$$

Crucially, Alice and Bob cannot compute QBER over their *entire* sifted key without destroying its secrecy — doing so would mean publicly revealing every bit. Instead, they publicly reveal and compare only a randomly chosen *subset* of the sifted key, then permanently discard those revealed bits (since they are now known to Eve) before proceeding. The

resulting QBER estimate is used to decide whether to proceed with the protocol at all: if QBER exceeds a security threshold (approximately 11% for BB84, under the security proofs discussed in Section 3.7), Alice and Bob abort, since above this threshold it becomes theoretically possible for an eavesdropper to possess the entire key while remaining statistically indistinguishable from ordinary noise.

3.5 Eavesdropping Detection

The canonical eavesdropping strategy analyzed for BB84 is the *intercept-resend attack*: Eve intercepts each transmitted qubit, measures it in a randomly guessed basis θ_E , and forwards a freshly prepared qubit — based on her own measurement outcome — onward to Bob. Since Eve does not know Alice’s true basis θ_A , she guesses correctly with probability $\frac{1}{2}$.

When Eve guesses correctly ($\theta_E = \theta_A$), her measurement reproduces Alice’s state exactly (per the matched-basis case of the Born rule, Section 2.4), and she forwards an identical, undisturbed qubit to Bob — introducing no error. When she guesses incorrectly ($\theta_E \neq \theta_A$), her measurement randomizes the state (the mismatched-basis case), and the qubit she forwards is no longer reliably correlated with Alice’s original bit; if Bob’s basis happens to match Alice’s original basis (the only case that survives sifting), his measurement outcome is now only correct with probability $\frac{1}{2}$, not certainty. Combining these cases:

$$\text{QBER}_{\text{intercept-resend}} = P(\theta_E \neq \theta_A) \times \frac{1}{2} = \frac{1}{2} \times \frac{1}{2} = \frac{1}{4} = 25\%$$

A QBER of 25% is far above the security threshold, making a full intercept-resend attack on every qubit easily detectable. More generally, an information-disturbance tradeoff — formalized in later security proofs — guarantees that *any* strategy Eve uses to extract partial information from a qubit necessarily introduces disturbance proportional to the information gained; there is no way for her to eavesdrop “for free,” a generalization beyond the single intercept-resend example that underlies the unconditional security proofs of Section 3.7.

3.6 Privacy Amplification and Key Distillation

After QBER is estimated and found acceptable, two further classical post-processing steps convert the noisy, partially-known sifted key into a short but provably secret final key.

Error correction (also called information reconciliation) allows Alice and Bob to correct the remaining discrepancies in their sifted keys, using protocols such as Cascade [5], which exchange parity information over the public channel in a manner designed to leak as little information as possible to Eve, while still allowing Bob to locate and fix his errors.

Privacy amplification [6] then compresses the now-identical corrected key using a randomly chosen universal hash function, deliberately shortening it. The number of bits removed is calculated from the estimated QBER, via the binary Shannon entropy function $H_2(p) = -p \log_2 p - (1-p) \log_2 (1-p)$. The resulting Secure Key Rate is:

$$R_{\text{SKR}} \approx R_{\text{sifted}} [1 - f_{\text{EC}} H_2(\text{QBER}) - H_2(\text{QBER})]$$

where $f_{EC} \geq 1$ is the error-correction inefficiency factor. This hashing step makes Eve’s fractional knowledge of the compressed key exponentially small, converting a merely *identical* key into a genuinely *secret* one.

3.7 Security Proofs and Their Assumptions

BB84 is described as possessing *unconditional* (information-theoretic) security: its guarantee does not depend on any assumption about Eve’s computational power, unlike classical schemes such as RSA, which rely on the presumed difficulty of certain mathematical problems. This distinction is fundamental — BB84’s security rests on the laws of physics (no-cloning and measurement disturbance) rather than on unproven computational hardness. The first fully rigorous proofs of this unconditional security came from later work, notably by Mayers [7], and in a notably clean and influential form by Shor and Preskill [8], who connected BB84’s security to quantum error-correcting codes.

This security, however, holds only under a specific set of assumptions: genuinely random basis and bit choices, an authenticated (though not necessarily secret) classical channel, and correctly-characterized devices matching the idealized theoretical model. Section 3.8 examines how real implementations can deviate from these idealized assumptions.

3.8 Real-World Limitations

Several practical effects complicate the idealized picture developed above, and are directly relevant to the noisy-simulator and real-hardware work presented in Part III of this report.

Channel loss. Real quantum channels (optical fiber or free space) attenuate signals exponentially with distance, characterized by a loss coefficient α (dB/km) such that transmittance $\eta = 10^{-\alpha L/10}$. Loss reduces the raw key generation rate but does not, by itself, introduce bit errors — a lost photon simply fails to arrive.

Finite-key effects. Any real QBER estimate is computed from a *finite* sample, and therefore carries genuine statistical uncertainty around the true underlying error rate. Rigorous finite-key security analysis [9] uses concentration inequalities to bound this uncertainty and requires sacrificing additional key bits as a conservative statistical margin — an effect examined quantitatively for this project’s own data in Chapter 9.

Decoy states. Practical BB84 implementations typically use attenuated laser pulses (weak coherent pulses) rather than true single-photon sources, since ideal single-photon sources remain technologically difficult. This introduces a small probability of multi-photon pulses, exploitable via a *photon-number-splitting* attack, in which Eve can extract a photon from a multi-photon pulse without disturbing the rest of the signal. The decoy-state method [10, 11] counters this by randomly varying pulse intensity and statistically comparing detection rates across intensities to detect such an attack.

Detector imperfections. Real single-photon detectors exhibit both a nonzero *dark count* rate (false detections with no incoming photon) and imperfect *detection efficiency* (failing to register some photons that do arrive). Dark counts introduce a noise floor on achievable QBER that grows relatively more significant at longer transmission distances, where the true signal rate is lower.

With BB84's full theoretical structure now established, Chapter 4 turns to LM05 — a protocol that achieves the same cryptographic goal through a structurally different, two-way mechanism.

Chapter 4

LM05 — Lucamarini and Mancini’s Two-Way Protocol

4.1 Historical Context and Motivation for a Deterministic Protocol

LM05 was introduced by Marco Lucamarini and Stefano Mancini in 2005, in a paper titled “Secure Deterministic Communication Without Entanglement,” published in *Physical Review Letters* [2]. The protocol’s name follows the same convention as BB84: the authors’ initials combined with the year of publication.

The word “deterministic” in the original paper’s title points to the central motivation behind LM05, and the key structural difference from BB84 that this chapter will develop in full: BB84, as shown in Chapter 3, discards roughly half of its raw transmitted qubits during basis reconciliation, purely due to the random mismatch between Alice’s and Bob’s independently chosen bases. LM05 was designed to avoid this loss by restructuring the protocol so that the party performing the final measurement always measures in a basis they themselves chose and remember — eliminating the need for a public basis-matching lottery, at the cost of introducing an entirely different structural element: a qubit that travels the channel twice instead of once.

4.2 The Two-Way Round-Trip Communication Model

LM05 involves the same two parties, Alice and Bob, and the same two channel types as BB84, but reverses and extends the flow of the quantum transmission into two distinct legs.

Leg 1 (Bob to Alice). Bob, not Alice, initiates each round. He randomly and independently chooses a basis $\theta_B \in \{Z, X\}$ and a bit $b \in \{0, 1\}$, prepares a qubit using exactly the same encoding rule as BB84’s Alice (Section 3.2):

$$|\psi\rangle_{\text{prepared}} = H^{[\theta_B=X]} X^{[b=1]} |0\rangle$$

and transmits this qubit to Alice over the quantum channel. Bob privately records his own choice of θ_B and b for later use — this record is essential, since he is the same party

who will perform the final measurement at the end of the round trip.

Leg 2 (Alice back to Bob, Message Mode only). As will be detailed in Section 4.3, in one of the two possible modes Alice applies an operation to the qubit and transmits it back to Bob, who then performs his final measurement using his own recorded basis θ_B from Leg 1 — not a newly chosen random basis. This reuse of Bob’s own basis is precisely what removes the need for public basis reconciliation: since Bob already knows the correct basis (his own), there is no lottery to resolve.

4.3 Control Mode and Message Mode

Upon receiving Bob’s qubit at the end of Leg 1, Alice makes a second independent random choice: a mode $m \in \{\text{CM}, \text{MM}\}$ (Control Mode or Message Mode), typically with equal probability, though the protocol does not strictly require a 50/50 split.

Control Mode serves purely as a security check. Alice measures the received qubit immediately, in a randomly and independently chosen basis $\theta_{A,\text{CM}} \in \{Z, X\}$, obtaining an outcome via the Born rule. She does not transmit anything back to Bob for this round; instead, she later publicly announces both her basis $\theta_{A,\text{CM}}$ and her measurement outcome. Bob compares this against his own recorded (θ_B, b) for that round, but only for the subset of Control Mode rounds where $\theta_{A,\text{CM}} = \theta_B$ happened to match — exactly analogous to the basis-matching condition from BB84, but restricted to only the Control Mode subset of rounds, and doubly filtered:

$$P(\text{CM and basis match}) = P(\text{CM}) \times P(\theta_{A,\text{CM}} = \theta_B) = \frac{1}{2} \times \frac{1}{2} = \frac{1}{4}$$

This $\frac{1}{4}$ probability, compared to BB84’s $\frac{1}{2}$ sifting probability, is the direct origin of a structural asymmetry examined quantitatively in Chapter 9: LM05’s QBER-testing sample size is, for a given number of raw qubits N , only about half the size of BB84’s equivalent sample.

Message Mode is where the actual key-carrying information is transmitted. Alice does not measure the qubit at all in this mode; instead, she applies a fixed operation encoding one classical bit, described fully in Section 4.4, and transmits the resulting qubit back to Bob as Leg 2 of the round trip. Since Message Mode involves no basis choice or comparison at all, every Message Mode round contributes directly and fully to the eventual sifted key — with no further discarding, in contrast to Control Mode’s doubly-filtered $\frac{1}{4}$ probability.

4.4 Deterministic Bit Encoding via Identity and Pauli-Y

In Message Mode, Alice encodes one bit by choosing between two fixed, basis-independent operations. To encode bit 0, she applies the identity operator I — leaving the qubit completely unchanged. To encode bit 1, she applies the Pauli-Y operator:

$$Y = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}$$

The choice of Y , rather than the more obvious bit-flip candidate X , is deliberate and mathematically necessary. Alice does not know Bob's original preparation basis θ_B when she applies her encoding operation, yet her operation must reliably flip the measurable outcome regardless of which basis Bob used. Computing the action of Y on all four basis states:

$$Y|0\rangle = i|1\rangle, \quad Y|1\rangle = -i|0\rangle, \quad Y|+\rangle = -i|-\rangle, \quad Y|-\rangle = i|+\rangle$$

In every case, the measurable basis state is flipped, and the accompanying phase factors (i or $-i$) have no effect on measurement probabilities, since $|i|^2 = |-i|^2 = 1$ and the Born rule depends only on the squared magnitude of the relevant amplitude. By contrast, applying X instead would fail silently for half of all rounds: $X|+\rangle = |+\rangle$, meaning X leaves the X-basis state $|+\rangle$ completely unchanged rather than flipping it — an encoding failure that would only manifest whenever Bob happened to have prepared in the X basis. This basis-universal flip property is the specific algebraic reason Y , and only Y , is the correct choice for LM05's basis-blind encoding requirement.

At the end of Leg 2, Bob measures the returning qubit in his own originally recorded basis θ_B , and decodes Alice's bit by comparing his outcome to his own originally prepared bit b :

$$\text{decoded bit} = \begin{cases} 0 & \text{if outcome} = b \quad (\text{Alice applied } I) \\ 1 & \text{if outcome} \neq b \quad (\text{Alice applied } Y) \end{cases}$$

Since Bob always measures in the basis he himself chose, this decoding is fully deterministic in the absence of noise or eavesdropping — precisely the property referenced in the protocol's original title.

4.5 QBER Estimation via Control Mode

LM05's QBER is defined analogously to BB84's, but computed exclusively from the basis-matched Control Mode subset described in Section 4.3:

$$\text{QBER}_{\text{LM05}} = \frac{\text{number of CM basis-matched disagreements}}{\text{total CM basis-matched rounds}}$$

A structurally important distinction from BB84 is worth emphasizing here: LM05 dedicates an entirely separate subset of rounds (Control Mode) purely to security testing, while Message Mode rounds are never used for QBER estimation and always contribute to the key. BB84, by contrast, must sacrifice a portion of its single sifted-key pool for QBER sampling, since it has no separate dedicated testing channel. This gives LM05 a cleaner separation between "testing" and "key material," though, as just derived, at the cost of a smaller absolute testing sample size for a given N .

4.6 The Lucamarini Attack and Known Vulnerabilities

The original 2005 paper's security argument addresses probabilistic intercept-resend-style attacks, directly analogous to the BB84 analysis of Section 3.5: an eavesdropper who inter-

cepts and measures a qubit in a randomly guessed basis introduces detectable disturbance, visible in Control Mode statistics.

However, a stronger attack exists against the unmodified protocol, documented in Lucamarini’s own doctoral thesis and related to independently discovered man-in-the-middle attacks on two-way QKD protocols more broadly [4]. Rather than intercepting and forwarding Bob’s genuine qubit, an eavesdropper Eve can instead intercept and discard it entirely, substituting her own freshly prepared “decoy” qubit — in a basis of her own choosing, which she therefore knows with certainty — and sending this decoy onward to Alice. Alice, unaware the qubit did not originate from Bob, performs her normal random Control Mode or Message Mode procedure on Eve’s decoy. In Control Mode, Eve can correctly reconstruct the expected statistics, since she knows her own decoy’s true state. In Message Mode, Eve intercepts the qubit returning from Alice (now carrying Alice’s operation applied to Eve’s own decoy state, not Bob’s), measures it in her known basis to read Alice’s bit with certainty, and forwards an appropriately re-encoded signal onward to Bob, mimicking what he would have expected from the legitimate protocol. Because Eve controls both directions of the channel for this qubit and always knows her decoy’s basis, this attack introduces *no* detectable disturbance in Control Mode statistics — a categorically more powerful attack than probabilistic intercept-resend, since it does not rely on guessing.

4.7 Security Proofs and Their Narrower Assumptions

The vulnerability described above demonstrates that LM05’s original disturbance-based security claim holds only against probabilistic intercept-resend-style attacks, not against a full channel-controlling man-in-the-middle adversary. Countermeasures in the broader two-way QKD literature include measurement-device-independent (MDI) protocol variants, which remove trust assumptions about measurement devices entirely, and strict optical isolation techniques to prevent the kind of channel substitution the attack above depends on.

This represents a genuinely narrower assumption set than BB84’s: LM05’s security proof requires not only the standard assumptions (true randomness, authenticated classical channel, correctly characterized devices) but also protection against full channel substitution — a threat model that does not arise at all in BB84’s simpler one-way structure. This distinction is revisited quantitatively in Chapter 5’s direct comparison of the two protocols’ security assumptions.

4.8 Real-World Limitations

Several of BB84’s practical limitations (Section 3.8) reappear for LM05 in modified, generally more severe form, due to the two-way channel structure.

Channel loss compounds multiplicatively across the two legs of the round trip: for a symmetric channel of one-way distance L and loss coefficient α , the round-trip transmittance is $\eta_{\text{round-trip}} = (10^{-\alpha L/10})^2 = 10^{-2\alpha L/10}$ — double the loss, in decibel terms, that BB84 experiences over the same one-way distance. This means LM05’s maximum practical transmission distance is generally shorter than BB84’s, despite LM05’s advantage in per-round

key efficiency established in Section 4.3.

Dual-detector exposure. LM05 involves two independent detectors — Alice’s (used in Control Mode) and Bob’s (used in Message Mode) — each contributing its own dark-count rate and detection inefficiency to different parts of the protocol’s statistics, in contrast to BB84’s single detector role for Bob.

Decoy-state complications. Photon-number-splitting attacks (Section 3.8) can, in principle, be attempted on either leg of LM05’s round trip. A split on the return leg is potentially more damaging than a split on the outbound leg, since Alice’s fixed I/Y encoding (Section 4.4) can be correctly identified by Eve in *any* measurement basis, without needing to guess Bob’s original preparation basis — an asymmetry that does not have a direct analogue in BB84’s simpler one-way decoy-state analysis, and would require a two-way-specific decoy-state scheme to fully address, an active area of ongoing research beyond the scope of this project’s implementation.

With both protocols now developed in full theoretical detail, Chapter 5 places them side by side for a direct structural and security comparison, setting up the empirical questions that Part III of this report investigates through implementation.

Chapter 5

BB84 and LM05 Side by Side — A Theoretical Comparison

With both protocols developed individually in Chapters 3 and 4, this chapter draws them together directly, examining four dimensions along which they differ, before Part III turns to testing whether these theoretical differences actually manifest in implemented circuits and measured data.

5.1 Structural Differences: One-Way Versus Two-Way

The most fundamental difference between BB84 and LM05 is the direction and number of quantum transmissions per round. In BB84, a single qubit travels once, from Alice to Bob, and the protocol’s entire logic — encoding, transmission, measurement, sifting — unfolds around this one-way journey. In LM05, a single qubit is involved in what is, at least in the idealized protocol description, two separate physical transits: outward from Bob to Alice, and then, for Message Mode rounds, back again from Alice to Bob.

This is not a superficial difference in bookkeeping; it changes which party initiates the round (Bob, in LM05, rather than Alice, as in BB84), it changes where the classical bit ultimately gets recorded (in Bob’s memory from the start, in LM05, versus freshly generated by Alice at the point of transmission, in BB84), and it introduces an entirely new category of security consideration — the possibility of an eavesdropper attacking either leg of the round trip independently, or, as shown in Section 4.6, exploiting the round-trip structure itself through a channel-substitution attack that has no equivalent in a purely one-way protocol.

It is worth being precise, in anticipation of Part III, about a caveat that will be raised again there: any circuit-level implementation of LM05 on a single quantum processor necessarily represents this two-way round trip as a sequence of gates applied within one circuit — state preparation, then an intermediate operation, then final measurement — rather than as a qubit physically leaving a chip, traveling somewhere, and returning. This is a faithful representation of the protocol’s *logical* structure, but not of a genuine physical two-hop channel journey; Chapter 11 returns to this distinction as one of the project’s explicitly stated limitations.

5.2 Efficiency Comparison: Sifting Loss Versus Deterministic Key Generation

BB84’s basis reconciliation step (Section 3.3) discards, on average, half of all raw transmitted qubits, purely due to the random mismatch between Alice’s and Bob’s independently chosen bases — a $\frac{1}{2}$ probability of survival per raw qubit. LM05’s Message Mode rounds, by contrast, suffer no equivalent basis-mismatch loss at all: because Bob always measures in the same basis he himself originally chose (Section 4.2), every Message Mode round deterministically contributes to the sifted key. The only loss LM05 experiences at the key-generation stage is the initial CM/MM split itself — typically also a $\frac{1}{2}$ probability, but for a qualitatively different reason (a deliberate security-testing allocation, not a matching failure).

Numerically, for N raw qubits, this gives:

$$\mathbb{E}[\text{BB84 sifted key}] = \frac{N}{2}, \quad \mathbb{E}[\text{LM05 Message Mode key}] = \frac{N}{2}$$

These two expectations are, perhaps surprisingly, numerically identical — both protocols convert roughly half of their raw qubits into usable key material, but for entirely different structural reasons. This near-parity is directly confirmed by this project’s own ideal-simulator results, presented in Chapter 7. Where the two protocols diverge sharply is not in raw key-generation efficiency, but in the size of the sample available for *security testing* (QBER estimation): BB84 can, in principle, draw its QBER sample from within its single sifted-key pool, while LM05’s QBER sample is restricted to the doubly-filtered Control Mode subset described in Section 4.5, occurring with probability only $\frac{1}{4}$ per raw qubit rather than $\frac{1}{2}$ — a genuine efficiency disadvantage for LM05 in the specific currency of statistical confidence, examined quantitatively in Chapter 9.

5.3 Security Assumption Comparison

Table 5.1 summarizes the security assumption differences established across Sections 3.7, 3.8, 4.6, and 4.7.

Aspect	BB84	LM05
Channel direction	One-way	Two-way (round trip)
Baseline attack model	Intercept-resend	Intercept-resend
Known stronger attack	None structurally equivalent	Channel-substitution man-in-the-middle (Section 4.6)
Additional required assumption	—	Protection against full channel control by an adversary
Security proof maturity	Extensive, decades of literature [7, 8]	Comparatively newer; MDI variants an active research area

The key conclusion from this comparison is that LM05’s security rests on a strictly narrower set of assumptions than BB84’s: everything BB84 requires, LM05 also requires, but LM05 additionally requires protection against a channel-substitution attack with no equivalent in BB84’s simpler one-way structure. This is not a claim that LM05 is insecure in practice — countermeasures exist, as noted in Section 4.7 — but it is a meaningful, quantifiable difference in the assumption burden each protocol places on its physical implementation.

5.4 Why LM05 Is Expected to Underperform BB84 Under Noise

The final theoretical comparison concerns not security assumptions but raw physical performance under realistic, noisy conditions — the question this entire project’s implementation work (Part III) is ultimately designed to test empirically.

The argument is straightforward once stated: any physical qubit transmission through a real, noisy channel or noisy gate accumulates some probability of error per transit. BB84 exposes its qubit to this accumulated error exactly once per round. LM05’s Message Mode rounds — the rounds that actually contribute to its key — expose the same qubit to accumulated error *twice*: once on the outbound leg from Bob to Alice, and once again on the return leg from Alice back to Bob. To a good first approximation, for small per-transit error probabilities ϵ , these two independent error contributions add:

$$\epsilon_{\text{LM05, MM round}} \approx \epsilon_{\text{outbound}} + \epsilon_{\text{return}} \approx 2\epsilon_{\text{BB84, one round}}$$

This is not merely a hand-waving argument: Chapter 6 will make it fully concrete at the level of individual gate counts, showing that LM05’s Message Mode circuit literally contains one additional gate operation relative to BB84’s circuit, tying this theoretical expectation directly to a countable feature of the implemented circuit design. The theoretical prediction that follows is unambiguous: under any realistic noise source — whether simulated (Chapter 8) or genuinely physical (Chapter 9) — LM05’s quantum bit error rate should be measurably higher than BB84’s, all else being equal. Whether this prediction survives contact with an actual noisy simulator and, ultimately, a real IBM quantum processor, is the central empirical question that the remainder of this report sets out to answer.

Part III

From Theory to Code

Chapter 6

The Mathematics of Circuit Design

Part II developed BB84 and LM05 as mathematical protocols, described in terms of abstract state vectors and idealized measurements. This chapter makes the transition from that abstract description to an actual quantum circuit — the concrete object that a quantum computer, whether simulated or real, can execute. Every gate used in this project’s implementation is defined here in full, together with an explanation of why that specific gate, and no other, was the correct choice for its role.

6.1 Gate Matrices: X, H, Y, I

Four gate matrices, all acting on a single qubit, are sufficient to build every circuit used in this project:

$$X = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad H = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}, \quad Y = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}, \quad I = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$$

The Pauli-X gate acts as a classical NOT gate on the computational basis, exchanging $|0\rangle \leftrightarrow |1\rangle$. The Hadamard gate H is a basis-changing operator, satisfying the important self-inverse property $H^2 = I$ (readily verified by direct matrix multiplication), which converts between the Z basis and X basis states of Chapter 2. The Pauli-Y gate combines a bit-flip with an imaginary phase, as will be used specifically in LM05’s Message Mode encoding. The identity I represents ”do nothing,” included here for completeness since it appears explicitly in LM05’s encoding rule (Section 4.4).

6.2 Why Each Gate Was Chosen

It is not enough to state what each gate does; the design of both protocols’ circuits depended on specific, non-interchangeable properties of each gate, and it is worth making this reasoning explicit.

The X gate for bit encoding. X was chosen because it is the simplest possible operation that toggles between the two computational basis states with perfect fidelity and no phase ambiguity, exactly matching the requirement of encoding a single classical bit into $|0\rangle$ or $|1\rangle$ starting from a circuit’s default initial state $|0\rangle$.

The H gate for basis selection. The Hadamard gate’s self-inverse property, $H^2 = I$, is what allows a single gate definition to serve two separate algorithmic roles. Applied once, during encoding, it converts a Z-basis state into the corresponding X-basis state. Applied again, before measurement, it rotates the state back into the Z basis — the only basis in which a real quantum processor’s native measurement operator can read out a result. Because $H^2 = I$ exactly, this ”undo” operation is mathematically perfect when both parties happen to have used the same basis (Section 3.3’s sifting condition), and it produces genuine, unbiased 50/50 randomization when the bases differ (Section 2.4’s mismatched-basis case) — precisely the two behaviors BB84’s security model requires from one gate.

The Y gate, specifically, for LM05’s Message Mode. This choice deserves particular attention, since it is the least obvious of the four. Alice, in Message Mode, must encode a bit using an operation that flips the measurable outcome regardless of which basis Bob used to prepare the qubit — information she does not have access to. The seemingly natural candidate, X , fails this requirement: while $X|0\rangle = |1\rangle$ correctly flips a Z-basis state, $X|+\rangle = |+\rangle$ leaves an X-basis state completely unchanged. Using X would therefore cause LM05’s Message Mode encoding to fail silently whenever Bob happened to have prepared in the X basis, corrupting roughly half of all Message Mode rounds. The Pauli-Y gate, as derived algebraically in Section 4.4, flips the measurable outcome in *both* bases simultaneously, with the resulting phase factors having no effect on measurement probabilities. This basis-universal flip property is the specific mathematical reason Y , and only Y , is the correct gate for this role.

The barrier instruction. In addition to the four gate matrices, both circuits make use of a Qiskit `barrier()` instruction, which carries no mathematical effect whatsoever on the qubit’s state — it is a pure compiler directive, preventing the circuit optimizer from merging or reordering gates across a conceptual boundary (in this case, the boundary between one party’s operations and the other’s). It is included for structural clarity in the circuit’s construction and diagrammatic representation, not because it changes any computed result.

6.3 Building the BB84 Circuit, Gate by Gate

The complete BB84 circuit for a single round, exactly as implemented, applies the following sequence of operations to a qubit initialized at $|0\rangle$:

$$|0\rangle \xrightarrow{X^{[a=1]}} \xrightarrow{H^{[\theta_A=X]}} \xrightarrow{\text{barrier}} \xrightarrow{H^{[\theta_B=X]}} \xrightarrow{\text{measure}} c_0$$

Concretely worked through for a specific case — Alice’s bit $a = 1$, basis $\theta_A = X$ — the encoding step proceeds as two explicit matrix-vector multiplications:

$$X|0\rangle = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} 1 \\ 0 \end{pmatrix} = \begin{pmatrix} 0 \\ 1 \end{pmatrix} = |1\rangle, \quad H|1\rangle = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix} \begin{pmatrix} 0 \\ 1 \end{pmatrix} = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ -1 \end{pmatrix} = |-\rangle$$

confirming, by direct computation, that bit=1 encoded in the X basis produces exactly $|-\rangle$, matching the general encoding rule of Section 3.2. If Bob’s basis θ_B also happens to be X (a matched-basis round), his pre-measurement gate applies H once more:

$$H|-\rangle = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix} \cdot \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ -1 \end{pmatrix} = \frac{1}{2} \begin{pmatrix} 1-1 \\ 1+1 \end{pmatrix} = \begin{pmatrix} 0 \\ 1 \end{pmatrix} = |1\rangle$$

and the Born rule then gives $P(\text{outcome} = 1) = |\langle 1|1\rangle|^2 = 1$ — a certain, correct recovery of Alice’s original bit. If instead Bob’s basis had been Z (a mismatched-basis round), no second H is applied, and the Born rule on the state $|-\rangle = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ -1 \end{pmatrix}$ measured directly in the computational basis gives $P(\text{outcome} = 0) = P(\text{outcome} = 1) = \frac{1}{2}$ — the random, uninformative result that basis reconciliation (Section 3.3) is designed to discard.

6.4 Building the LM05 Circuit, Gate by Gate

LM05’s circuit construction begins identically to BB84’s (Bob now playing the role of preparer), but then diverges into two mutually exclusive branches depending on Alice’s randomly chosen mode.

Bob’s preparation, common to both branches:

$$|0\rangle \xrightarrow{X^{[b=1]}} \xrightarrow{H^{[\theta_B=X]}} \xrightarrow{\text{barrier}} |\psi\rangle_{\text{prepared}}$$

Control Mode branch:

$$|\psi\rangle_{\text{prepared}} \xrightarrow{H^{[\theta_{A,CM=X}]}} \xrightarrow{\text{measure}} c_0$$

Message Mode branch:

$$|\psi\rangle_{\text{prepared}} \xrightarrow{Y^{[\text{msg}=1]}} \xrightarrow{H^{[\theta_B=X]}} \xrightarrow{\text{measure}} c_0$$

Worked through for a concrete Message Mode example — Bob prepares bit $b = 0$ in the Z basis (so $|\psi\rangle_{\text{prepared}} = |0\rangle$), and Alice wishes to encode message bit 1:

$$Y|0\rangle = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix} \begin{pmatrix} 1 \\ 0 \end{pmatrix} = \begin{pmatrix} 0 \\ i \end{pmatrix} = i|1\rangle$$

Bob’s final measurement gate, since his original basis was Z, applies no further H :

$$P(\text{outcome} = 0) = |\langle 0|(i|1\rangle)|^2 = |i \cdot 0|^2 = 0, \quad P(\text{outcome} = 1) = |\langle 1|(i|1\rangle)|^2 = |i \cdot 1|^2 = |i|^2 = 1$$

Outcome 1 is obtained with certainty; note that the phase factor i appears throughout the intermediate calculation and vanishes completely in the final probability, exactly as the general argument of Section 4.4 predicted. Bob then decodes: outcome (1) $\neq$ his original bit (0), so he correctly decodes Alice’s message bit as 1.

6.5 The Complete Circuit Signature for Both Protocols

Collecting the gate sequences derived above, the maximum-depth circuit for each protocol branch is:

Circuit	Maximum gate sequence
BB84, one round	$X \rightarrow H \rightarrow H \rightarrow \text{measure}$ (3 conditional gates)
LM05, Control Mode round	$X \rightarrow H \rightarrow H \rightarrow \text{measure}$ (3 conditional gates)
LM05, Message Mode round	$X \rightarrow H \rightarrow Y \rightarrow H \rightarrow \text{measure}$ (4 conditional gates)

This table makes the theoretical prediction of Section 5.4 numerically concrete: LM05’s Message Mode branch — the only branch that actually contributes to its final key — contains one additional gate relative to BB84’s circuit or LM05’s own Control Mode branch. Since each additional gate carries its own nonzero physical error probability, this single extra gate is the precise, countable origin of LM05’s expected higher quantum bit error rate under any realistic noise source, a prediction that Chapters 8 and 9 test directly against simulated and real quantum hardware data respectively.

With circuit design now fully specified at the gate level, Chapter 7 turns to the first stage of actual implementation: running these circuits on a noise-free simulator to validate that the circuit logic itself, independent of any physical imperfection, behaves exactly as this chapter’s mathematics predicts.

Chapter 7

Level 1 — The Ideal Simulator

7.1 Why Validate on a Noise-Free Simulator First

Before either protocol was tested against realistic noise (Chapter 8) or genuine quantum hardware (Chapter 9), it was run on Qiskit Aer’s `AerSimulator()` with no noise model attached — a perfect, noise-free simulation in which every gate behaves as an exact unitary matrix multiplication and every measurement follows the Born rule exactly, with no probabilistic deviation beyond what the protocol’s own randomized bit and basis choices introduce.

The purpose of this stage is methodological rather than scientific: it isolates and validates the correctness of the *circuit logic itself*, independent of any question about physical noise. If a bug existed in the encoding, sifting, or decoding logic — for instance, an incorrectly applied gate, a basis comparison implemented backwards, or a decoding rule with an off-by-one error — it would very likely produce a nonzero QBER even in this idealized, noise-free setting, since the underlying mathematics of Chapter 6 predicts a QBER of exactly zero when the logic is correct. Confirming this zero-QBER prediction empirically, before introducing any noise, therefore serves as a necessary correctness check: any noise or error observed in later chapters can then be confidently attributed to the noise source being modeled, rather than to an error in the implementation itself.

7.2 Implementation Walkthrough

For BB84, $N = 200$ raw qubits were generated, with Alice’s bits and both parties’ bases drawn independently and uniformly at random, following exactly the encoding and measurement procedure derived in Sections 3.2 and 6.3. Each round was built as an individual circuit and executed with a single shot on `AerSimulator()`, following which the sifting condition of Section 3.3 was applied, and QBER was computed over the resulting sifted key.

For LM05, the same $N = 200$ raw qubits were used for Bob’s preparation, with Alice’s mode, Control Mode basis, and Message Mode bit likewise drawn independently and uniformly at random, following the branching circuit structure derived in Sections 4.2–4.4 and 6.4. QBER was computed from the basis-matched Control Mode subset (Section 4.5), while the sifted key was built exclusively from Message Mode rounds, with Bob’s decoding rule

(Section 4.4) applied to recover each bit.

7.3 Results and Their Exact Mathematical Explanation

The BB84 run produced a sifted key of length 100, a QBER of exactly 0.0000, and a Secure Key Rate fraction of exactly 1.0000 (100.0 secure key bits). The LM05 run produced 55 basis-matched Control Mode rounds with zero mismatches (QBER = 0.0000), a Message Mode sifted key of length 97, and likewise an SKR fraction of exactly 1.0000 (97.0 secure key bits). Both protocols additionally confirmed that Alice’s and Bob’s final keys matched bit-for-bit.

Each of these numbers admits a precise mathematical explanation, rather than being merely an observed coincidence.

The sifted key lengths. Since Alice’s and Bob’s basis choices in BB84 are independent uniform binary random variables, the number of surviving sifted bits K follows $K \sim \text{Binomial}(N = 200, p = 0.5)$, with $\mathbb{E}[K] = 100$ and standard deviation $\sigma = \sqrt{200 \times 0.5 \times 0.5} \approx 7.07$. The observed value of exactly 100 sits precisely at the distribution’s mean. For LM05’s Control Mode sample, the relevant probability is the doubly-filtered $p = 0.25$ derived in Section 4.3, giving $\mathbb{E} = 50$ and $\sigma = \sqrt{200 \times 0.25 \times 0.75} \approx 6.12$; the observed value of 55 lies comfortably within one standard deviation. For LM05’s Message Mode sifted key, the relevant probability reverts to $p = 0.5$ (Alice’s mode choice alone, with no further basis-matching filter, per Section 4.3), giving the same $\mathbb{E} = 100, \sigma \approx 7.07$ as BB84; the observed value of 97 again lies within one standard deviation. These are, in short, unremarkable, statistically ordinary outcomes of the binomial processes that Chapters 3 and 4 predict — confirming that the random number generation and mode/basis selection logic are behaving exactly as the underlying probability model specifies.

Why QBER is exactly zero, not merely close to zero. This result requires a qualitatively different explanation from the key-length results above, since it is not a statistical outcome at all — it is a deterministic consequence of the Born rule. As derived explicitly in Section 6.3, when Alice’s and Bob’s bases match (or, for LM05, when Alice’s Control Mode basis matches Bob’s preparation basis), the qubit’s state prior to measurement is guaranteed, by the algebra of the H gate’s self-inverse property, to be exactly an eigenstate of the measurement operator. The Born rule then assigns probability exactly 1 to the correct outcome and exactly 0 to the incorrect one — there is no probabilistic spread whatsoever in this idealized, noise-free setting. A QBER of 0.0000 is therefore not a fortunate simulation result; it is the only possible outcome the mathematics permits, for every single run of the ideal simulator, regardless of how many times the experiment is repeated.

Why the Secure Key Rate fraction is exactly 1.0000. Recall from Section 3.6 that $R_{\text{SKR}} = R_{\text{sifted}}[1 - f_{\text{EC}}H_2(\text{QBER}) - H_2(\text{QBER})]$, where H_2 is the binary Shannon entropy function. This function has the defining boundary property $H_2(0) = 0$ (by the standard convention that $\lim_{p \rightarrow 0} p \log_2 p = 0$). Substituting QBER = 0 directly gives $R_{\text{SKR}} = R_{\text{sifted}}[1 - 0 - 0] = R_{\text{sifted}}$ — the entire sifted key survives as secret key material, with nothing sacrificed to error correction (since there are no errors to correct) or privacy amplification

(since there is no error-derived uncertainty for an eavesdropper to exploit). This is precisely why the printed SKR fraction is exactly 1.0000 in both protocols' Level 1 output, and why the final secure key length exactly equals the sifted key length in each case.

7.4 What Zero QBER Proves — and Does Not Prove

It is worth being precise about the scope of what this chapter's results actually demonstrate, in keeping with the honest-accounting principle this report follows throughout (formalized further in Chapter 11). A QBER of exactly zero on the ideal simulator confirms that the circuit logic — the gate sequences, the sifting condition, and the decoding rule — are implemented correctly and behave exactly as the mathematics of Chapters 3, 4, and 6 predicts. It does *not* yet demonstrate anything about how either protocol behaves under realistic physical imperfection, since the ideal simulator by construction contains no noise source of any kind. In particular, the theoretical prediction from Section 5.4 — that LM05 should exhibit a higher QBER than BB84 due to its two-way channel exposure — cannot be tested at all at this stage, since both protocols trivially achieve zero QBER regardless of any structural difference between them, once noise is entirely absent. Testing that specific prediction is precisely the task Chapter 8 takes up next, by deliberately introducing a realistic noise model into the same circuits validated here.

Chapter 8

Level 2 — The Noisy Simulator

8.1 What a Noise Model Represents Physically

Chapter 7 validated circuit logic under conditions no real quantum processor can achieve: perfectly noiseless gates and perfectly faithful measurement. This chapter introduces a `NoiseModel` object into the same `AerSimulator`, altering the simulation in two specific, physically motivated ways.

Gate error is modeled as a depolarizing channel applied after each noisy gate, described by the map

$$\mathcal{E}(\rho) = (1 - p) \rho + p \frac{I}{2}$$

where ρ is the qubit’s density matrix and p is a calibrated error probability specific to that gate and qubit. This mixes a small amount of the maximally mixed state into the intended state, degrading an otherwise pure encoded state.

Readout error is modeled independently as a stochastic bit-flip applied directly to the classical measurement outcome, described by a confusion matrix

$$M = \begin{pmatrix} P(0|0) & P(0|1) \\ P(1|0) & P(1|1) \end{pmatrix}$$

Because the circuits used in this project are extremely shallow — at most three or four conditional gates deep, as established in Section 6.5 — readout error, together with the single gate error each qubit is exposed to, dominates the observed QBER; accumulated multi-gate depolarization plays a comparatively minor role at this circuit depth.

8.2 FakeManilaV2 and FakeLimaV2: What They Are and Their Limitations

The two noise models used in this chapter, `FakeManilaV2` and `FakeLimaV2`, are provided by Qiskit’s fake-backend infrastructure (detailed further in Appendix A). Each is constructed by capturing a frozen calibration snapshot — gate error rates, readout confusion matrices, and coherence times — from an actual IBM quantum device (`ibmq_manila` and `ibmq_lima`

respectively) at some point in the past, and packaging that snapshot for use in a local, noise-injected simulation.

This construction carries an important limitation, directly relevant to the comparison drawn in Section 8.5: because the calibration snapshot is frozen at the time it was captured, it does not reflect any subsequent recalibration, hardware improvement, or drift in the real device it was derived from. A prediction generated from a fake backend is therefore a prediction about *that specific historical calibration state*, not a live forecast of current device behavior — a distinction whose consequences become apparent when this chapter’s predictions are compared against genuinely live hardware data in Chapter 9.

8.3 Implementation Walkthrough

The same $N = 200$ raw qubits, generated identically to Chapter 7’s procedure, were run through both protocols’ circuits, now executed on `AerSimulator(noise_model=...)` rather than the noise-free simulator, once for each of the two fake backends. All other logic — encoding, sifting, Control Mode/Message Mode branching, QBER computation, and secure key rate calculation — was left entirely unchanged from Chapter 7, ensuring that noise model was the only variable distinguishing this chapter’s results from the previous one.

8.4 Results and Their Exact Mathematical Explanation

Noise Model	BB84 QBER	LM05 QBER	BB84 SKR (bits)	LM05 SKR (bits)
FakeManilaV2	0.0500	0.0652	38.1	27.6
FakeLimaV2	0.0800	0.0652	13.1	27.6

BB84’s sifted key length was 100 for both noise models; LM05’s Control Mode checkable sample was 46 for both, with a Message Mode sifted key of approximately 111.

Why BB84’s QBER differs between the two backends. With QBER no longer forced to zero, the number of mismatches M among $n = 100$ sifted bits follows $M \sim \text{Binomial}(n = 100, p = \epsilon)$, where ϵ is the combined gate-and-readout error probability. The observed mismatch counts are $M_{\text{Manila}} = 0.05 \times 100 = 5$ and $M_{\text{Lima}} = 0.08 \times 100 = 8$. This difference reflects a genuine difference in the underlying calibration snapshots — FakeLimaV2’s captured device state has a higher combined error rate than FakeManilaV2’s for the relevant qubits. The binomial standard deviation at the pooled average $\epsilon \approx 0.065$ is $\sigma = \sqrt{100 \times 0.065 \times 0.935} \approx 2.47$ mismatches, meaning the observed gap of three mismatches is within roughly 1.2σ of pure sampling variation — consistent with, though not conclusively proving, a genuine underlying calibration difference between the two devices.

Why LM05’s QBER is identical across both backends. LM05’s QBER of 0.0652 corresponds to exactly $0.0652 \times 46 = 3$ mismatches out of 46 checkable rounds, for *both* noise models — an exact coincidence in raw mismatch count. Given the small sample size, the achievable QBER values are coarsely spaced ($2/46 = 0.0435$, $3/46 = 0.0652$, $4/46 = 0.0870$,

and so on), and the binomial standard deviation at $p \approx 0.065$ is $\sigma = \sqrt{46 \times 0.065 \times 0.935} \approx 1.67$ mismatches — meaning outcomes of 2, 3, or 4 mismatches are all within roughly one standard deviation of each other. Landing on exactly 3 mismatches for both backends is therefore a plausible coincidence at this sample size, not evidence that the two backends behave identically for LM05 specifically. This is a direct, numerical illustration of the finite-key sample-size concern raised theoretically in Section 5.2: LM05’s QBER estimate here rests on fewer than half the sample size BB84’s does.

The Secure Key Rate calculations, worked in full. For BB84 on FakeManilaV2 (QBER= 0.05, $n = 100$):

$$H_2(0.05) = -0.05 \log_2(0.05) - 0.95 \log_2(0.95) = 0.2161 + 0.0703 = 0.2863$$

$$R_{\text{SKR}} = 100 [1 - 1.16(0.2863) - 0.2863] = 100(0.3816) = 38.16 \approx 38.1 \text{ bits}$$

For BB84 on FakeLimaV2 (QBER= 0.08):

$$H_2(0.08) = -0.08 \log_2(0.08) - 0.92 \log_2(0.92) = 0.2915 + 0.1107 = 0.4022$$

$$R_{\text{SKR}} = 100 [1 - 1.16(0.4022) - 0.4022] = 100(0.1312) = 13.12 \approx 13.1 \text{ bits}$$

Note how sharply the secure key rate fell for only a three-percentage-point increase in QBER (5% to 8%): from 38.1 to 13.1 bits, nearly a third of its original value. This is the direct consequence of the binary entropy function’s steep curvature, and is precisely why practical QKD systems enforce a hard abort threshold (Section 3.4’s approximately 11% BB84 cutoff) rather than tolerating gracefully degrading security.

For LM05 (QBER= 0.0652, Message Mode key length $n \approx 111$):

$$H_2(0.0652) = 0.2569 + 0.0909 = 0.3478, \quad R_{\text{SKR}} = 111 [1 - 1.16(0.3478) - 0.3478] = 111(0.2487) \approx 27.6 \text{ bits}$$

8.5 The Prediction for Real Hardware

Averaging across both noise models gives an average QBER of 0.0650 for BB84 and 0.0652 for LM05 — LM05 is technically higher, consistent with the direction predicted theoretically in Section 5.4, but the margin (0.0002) is far smaller than the sampling noise floor computed above (on the order of ± 1.67 mismatches out of 46, corresponding to roughly $\pm 3.6\%$ in QBER terms). At this sample size, the predicted LM05-versus-BB84 QBER difference is not yet statistically distinguishable from zero — it is directionally consistent with theory, but not confirmed by this chapter’s data alone.

This chapter’s overall prediction for Chapter 9’s real-hardware experiment is therefore twofold: first, that BB84’s QBER should fall somewhere in the range observed here (roughly 5% to 8%) if the target hardware’s calibration resembles the historical FakeManilaV2/FakeLimaV2 snapshots; and second, that confirming LM05’s theoretically expected disadvantage will likely require a substantially larger raw qubit count than $N = 200$, in order to shrink the sampling noise floor enough for a genuine effect — if present — to become statistically visible above it. Chapter 9 tests both of these predictions directly against a live IBM quantum processor.

Chapter 9

Level 3 — Real IBM Quantum Hardware

9.1 IBM Quantum Platform and the Open Plan

The final and most significant escalation in this project’s implementation was to move beyond simulation entirely, submitting the same circuits validated in Chapters 7 and 8 to an actual physical quantum processor. This was made possible through IBM Quantum Platform [12], IBM’s cloud service for remote access to real superconducting quantum processors, accessed here through its free *Open Plan*, which provides a limited monthly allocation of genuine QPU compute time at no cost, shared across a global queue of other users on the same tier.

Access to the platform required creating an IBM Quantum account, generating an API token, and identifying the account’s Cloud Resource Name (CRN), all of which were used to authenticate a local `QiskitRuntimeService` session (further detail on this authentication workflow is given in Appendix A). Once authenticated, the account had access to several real backends, including `ibm_fez`, a 156-qubit processor, which was selected for this project’s experiments on the basis of having the shortest job queue among the available options at the time of submission.

9.2 Transpilation and Native Gate Sets

A circuit written using the abstract gates of Chapter 6 — X , H , Y — cannot be executed directly on real hardware. Every physical quantum processor supports only a specific, fixed set of *native gates*, dictated by its underlying physical control electronics; for `ibm_fez`, this native set consists of `cz`, `id`, `rz`, `sx`, and `x`. Before submission, every circuit must therefore be passed through a *transpiler* (implemented here via Qiskit’s `generate_preset_pass_manager`), which decomposes each abstract gate into an equivalent sequence of native gates. The Hadamard gate, for instance, decomposes as:

$$H = e^{i\pi/2} R_z(\pi/2) \sqrt{X} R_z(\pi/2)$$

This decomposition matters mathematically, not merely as an implementation detail: each native gate application carries its own calibrated error probability, and if a logical gate

decomposes into k native gates, the effective error probability compounds approximately as $\epsilon_{\text{logical}} \approx \sum_{i=1}^k \epsilon_i$ for small, independent per-gate errors. This is the precise physical mechanism underlying the theoretical prediction of Section 5.4 and the gate-count argument of Section 6.5: LM05’s Message Mode circuit, containing more logical gates than BB84’s, decomposes into a correspondingly larger number of native gate operations, and therefore accumulates more compounded native-gate error per round.

9.3 Implementation Walkthrough

Three separate experimental runs were conducted, at successively larger raw qubit counts, in order to investigate the effect of sample size on the reliability of the resulting QBER estimates — a concern raised theoretically in Section 3.8 and Section 8.5.

Run 1 ($N = 20$). An initial small-scale run was performed first, primarily to validate that the full pipeline — circuit construction, transpilation, job submission, queueing, and result retrieval — functioned correctly end to end before committing larger amounts of queue time to bigger runs.

Run 2 ($N = 200$). A second run, matching the raw qubit count used throughout Chapters 7 and 8, was performed to allow a like-for-like comparison against the simulator-based results of those chapters.

Run 3 ($N = 500$). A final, larger run was performed specifically to address the finite-key concern identified in Run 2’s results (detailed in Section 9.4 below), by substantially increasing the sample size available for QBER estimation, particularly for LM05’s doubly-filtered Control Mode subset.

In every run, both protocols’ circuits were submitted to the same backend (`ibm_fez`) on the same day, with the same shot count per circuit, specifically to avoid the calibration-drift bias that would arise from comparing results gathered under different, unrecorded hardware conditions — a methodological safeguard motivated directly by the channel-loss and detector-imperfection discussions of Sections 3.8 and 4.8.

9.4 Results Across Three Sample Sizes

Run	Protocol	N	Sample used	Mismatches	QBER
1	BB84	20	9 (sifted)	0	0.0000
1	LM05	20	8 (CM matched)	0	0.0000
2	BB84	200	105 (sifted)	2	0.0190
2	LM05	200	51 (CM matched)	0	0.0000
3	BB84	500	242 (sifted)	1	0.0041
3	LM05	500	127 (CM matched)	1	0.0079

Two immediate observations follow directly from Section 4.3’s earlier derivation. First, LM05’s checkable sample size in every run is noticeably smaller than BB84’s sifted key at the same N — for instance, 127 versus 242 at $N = 500$ — consistent with the theoretical $\mathbb{E}[\text{LM05 CM-matched}] = N/4$ compared to $\mathbb{E}[\text{BB84 sifted}] = N/2$ derived in Chapter 4.

Second, the raw mismatch counts throughout this table are extremely small — at most two in any single run — meaning that considerable care is required before drawing conclusions directly from the point-estimate QBER values alone, a concern addressed rigorously in the next section.

9.5 Statistical Analysis: Confidence Intervals and Significance Testing

The Run 2 result for LM05 — a QBER of exactly 0.0000 against BB84’s 0.0190 — appears, at first glance, to contradict the theoretical prediction of Section 5.4 that LM05 should exhibit a *higher* QBER than BB84. This apparent contradiction does not withstand a proper statistical treatment of the small sample size involved. If LM05’s true underlying error rate were, in fact, comparable to BB84’s observed 0.019, the probability of observing zero mismatches by chance alone, given only $n = 51$ trials, is:

$$P(0 \text{ mismatches} \mid n = 51, p = 0.019) = (1 - 0.019)^{51} \approx 0.373$$

A 37% probability of this exact outcome occurring purely by chance is far too large to treat the Run 2 result as meaningful evidence against the theoretical prediction; it is, instead, a textbook illustration of the finite-key sampling concern raised in Section 3.8.

A more rigorous treatment of the project’s best available data — Run 3, with the largest sample sizes — uses the Wilson score confidence interval, which remains reliable even for small mismatch counts (unlike the naive $\hat{p} \pm \sqrt{\hat{p}(1-\hat{p})/n}$ approximation, which can become unstable near zero):

$$p \in \frac{\hat{p} + \frac{z^2}{2n} \pm z \sqrt{\frac{\hat{p}(1-\hat{p})}{n} + \frac{z^2}{4n^2}}}{1 + \frac{z^2}{n}}, \quad z = 1.96 \text{ (95\% confidence)}$$

For BB84, Run 3 ($x = 1$, $n = 242$, $\hat{p} = 0.00413$):

$$p_{\text{BB84}} \in [0.00073, 0.02303] \quad (0.07\% \text{ to } 2.30\%)$$

For LM05, Run 3 ($x = 1$, $n = 127$, $\hat{p} = 0.00787$):

$$p_{\text{LM05}} \in [0.00139, 0.04326] \quad (0.14\% \text{ to } 4.33\%)$$

These two 95% confidence intervals overlap almost entirely. To test the difference formally, a two-proportion z-test was applied, pooling both mismatch counts under the null hypothesis that both protocols share the same true error rate:

$$\hat{p} = \frac{X_{\text{LM05}} + X_{\text{BB84}}}{n_{\text{LM05}} + n_{\text{BB84}}} = \frac{1 + 1}{127 + 242} = \frac{2}{369} \approx 0.00542$$

$$z = \frac{\hat{p}_{\text{LM05}} - \hat{p}_{\text{BB84}}}{\sqrt{\hat{p}(1-\hat{p}) \left(\frac{1}{n_{\text{LM05}}} + \frac{1}{n_{\text{BB84}}} \right)}} = \frac{0.00787 - 0.00413}{\sqrt{0.00542(0.99458) \left(\frac{1}{127} + \frac{1}{242} \right)}} \approx \frac{0.00374}{0.00805} \approx 0.464$$

A z -score of 0.464 corresponds to a two-tailed p -value of approximately 0.64 — far above any conventional significance threshold (such as 0.05). The honest statistical conclusion is therefore precise: the observed difference between BB84’s and LM05’s QBER at Run 3 is directionally consistent with the theoretical prediction of Section 5.4, but is *not* statistically significant at this sample size. Confirming the predicted effect with statistical confidence would require a substantially larger N than 500 — a specific, quantifiable next step discussed further in Chapter 12.

9.6 Comparing Real Hardware Against the Level 2 Prediction

Chapter 8 predicted a QBER range of approximately 2.9% to 3.9% for BB84 and 4.8% to 7.1% for LM05, based on the FakeManilaV2 and FakeLimaV2 noise models. The Run 3 real-hardware results (0.41% for BB84, 0.79% for LM05) fall substantially *below* both predicted ranges.

This discrepancy is fully explained by the limitation of fake-backend noise models identified in Section 8.2: FakeManilaV2 and FakeLimaV2 represent frozen historical calibration snapshots of older, smaller (5-qubit) devices, while `ibm_fez` is a newer, larger, and evidently better-calibrated 156-qubit processor whose live error rates are simply lower than the historical snapshots used for the Chapter 8 prediction. This is not a failure of the Level 2 methodology — the noise-injection mechanism itself was confirmed to work correctly in Chapter 8 — but a demonstration that a fake-backend noise model should be understood as a prediction *about a specific historical calibration state*, not a live forecast of any particular current device’s behavior, a distinction worth carrying forward into any future extension of this project that seeks tighter agreement between simulated prediction and real-hardware measurement.

With all three levels of implementation now complete and individually analyzed, Chapter 10 draws them together into a single combined comparison, synthesizing the mathematical progression from deterministic model to live physical measurement that this project has traced from Chapter 7 through this chapter.

Part IV

Synthesis

Chapter 10

Combined Analysis Across All Three Levels

10.1 The Mathematical Progression: Deterministic to Unknown-Stochastic

Considered together, Chapters 7 through 9 trace a specific and deliberate progression in the mathematical structure governing QBER, rather than three arbitrary, unrelated experiments. Chapter 7 (Level 1) is governed entirely by the Born rule with no noise operator present, giving $\text{QBER} \equiv 0$ deterministically on every run — there is no random variable to estimate at all. Chapter 8 (Level 2) introduces a *fixed, known* error probability ϵ , taken from a calibration snapshot, converting QBER into a genuine Binomial random variable $X \sim \text{Binomial}(n, \epsilon)$ whose parameter ϵ is nonetheless known in advance, by construction. Chapter 9 (Level 3) retains this same Binomial structure, but now ϵ is an *unknown*, live, physically real parameter that the experiment itself is attempting to estimate — a conceptual shift from simulation, where the experimenter controls and knows the noise parameter, to genuine experiment, where that parameter is the object of investigation.

This progression — deterministic model, to known-parameter stochastic model, to unknown-parameter stochastic model — is the precise reason a three-level implementation structure was necessary in the first place, rather than proceeding directly to real hardware. It is also the reason Chapter 9 required confidence intervals and significance testing while Chapter 7 required neither: the mathematical object being estimated only becomes genuinely uncertain, in the formal statistical sense, once the true underlying parameter is unknown.

10.2 QBER Trends Across Levels

	Level 1 (Ideal)	Level 2 (Noisy Sim.)	Level 3 (Real Hardware)
BB84 QBER	0.0000	0.0500 / 0.0800	0.0000 / 0.0190 / 0.0041
LM05 QBER	0.0000	0.0652 / 0.0652	0.0000 / 0.0000 / 0.0079

Two distinct transitions are visible in this table, and it is worth being precise about

what each one actually demonstrates. The transition from Level 1 to Level 2 is a *controlled* jump: a fixed noise model was deliberately injected into the simulator, so observing QBER rise from exactly zero to the 5–8% (BB84) and 6.5% (LM05) range confirms only that the noise-injection mechanism itself functions correctly — it is a validation of the implementation, not a discovery about physical reality. The transition from Level 2 to Level 3, by contrast, is the genuinely informative comparison: Chapter 8’s prediction assumed the FakeManilaV2/FakeLimaV2 calibration snapshots were a reasonable stand-in for real hardware behavior, and Level 3 tested this assumption directly. The real QBER values obtained (0.41%–1.9% for BB84, 0%–0.79% for LM05) came in substantially lower than the Level 2 prediction range, which Section 9.6 explained by the frozen, historical nature of the fake-backend calibration snapshots relative to the live, more modern `ibm_fez` device actually used.

It is also worth confronting directly the one data point that appears, superficially, to contradict the theoretical prediction of Chapter 5: Run 2’s LM05 result of 0.0000 against BB84’s 0.0190. As demonstrated rigorously in Section 9.5, this single data point carries a 37% probability of occurring purely by chance even if LM05’s true error rate matched BB84’s exactly, and should therefore be treated as statistically uninformative rather than as genuine evidence against the theoretical prediction. Of the three real-hardware comparisons available, the most statistically trustworthy — Run 3, with the largest sample sizes — is also the one that agrees with the theoretical direction (LM05 QBER exceeding BB84’s), though, as Section 9.5 established, not yet with formal statistical significance.

10.3 Secure Key Rate Trends Across Levels

Level	BB84 SKR (bits)	LM05 SKR (bits)	BB84 rate	LM05 rate
1 (ideal, $N = 200$)	100.0	97.0	50.0%	48.5%
2 (Manila, $N = 200$)	38.1	27.6	19.1%	13.8%
2 (Lima, $N = 200$)	13.1	27.6	6.6%	13.8%
3 (real, $N = 500$)	221.9	211.5	44.4%	42.3%

The Level 3 SKR values (221.9 and 211.5 bits, computed using the same Shannon-entropy formula of Section 3.6 applied to the Run 3 QBER values of Section 9.4) were not reported explicitly in Chapter 9’s results table but follow directly from the same procedure worked through in full in Section 8.4: for BB84, $H_2(0.0041) = 0.0384$, giving $R_{\text{SKR}} = 242[1 - 1.16(0.0384) - 0.0384] \approx 221.9$ bits; for LM05, $H_2(0.0079) = 0.0665$, giving $R_{\text{SKR}} = 247[1 - 1.16(0.0665) - 0.0665] \approx 211.5$ bits.

Two observations follow from this table. First, Level 1’s key rates (50.0% versus 48.5%) are nearly identical between the two protocols — direct numerical confirmation of the efficiency-parity argument developed in Section 5.2: absent any noise, LM05’s lack of basis-reconciliation loss compensates almost exactly for its CM/MM split, bringing its raw key-generation efficiency essentially in line with BB84’s, despite the two protocols achieving this efficiency through entirely different mechanisms. Second, Level 3’s key rates (44.4% versus 42.3%) sit much closer to Level 1’s near-ideal figures than to Level 2’s noticeably degraded

ones — reinforcing, from the SKR perspective, the same conclusion reached in Section 9.6 from the QBER perspective: `ibm_fez`’s real calibration is considerably closer to ideal than the historical Fake-backend noise profiles used to generate the Level 2 prediction.

10.4 Was the Theoretical Prediction Confirmed?

The central theoretical prediction of this report, established in Section 5.4 and made numerically concrete in Section 6.5 — that LM05 should exhibit a higher QBER than BB84 under realistic noise, due to its Message Mode circuit’s additional gate and its two-way channel’s compounded exposure to error — was tested at every level of this project’s implementation where noise was present.

At Level 2, the prediction held in direction (average LM05 QBER of 0.0652 versus BB84’s 0.0650) but not with any statistical force, the margin being far smaller than the sampling noise inherent to the sample sizes used. At Level 3, across three independent real-hardware runs, the prediction held in direction in two of three cases (Runs 1 and 3) and appeared, superficially, to fail in one case (Run 2) — an apparent failure that Section 9.5 demonstrated to be a statistically unsurprising consequence of that run’s small sample size, rather than genuine evidence against the theory. The most statistically robust real-hardware comparison available, Run 3’s two-proportion z-test, found the predicted direction present but not reaching conventional statistical significance ($p \approx 0.64$).

The honest, precise answer to this section’s title question is therefore neither a simple “yes” nor “no”: the prediction is *directionally consistent* with every piece of noisy data gathered across both simulation and real hardware, but has not yet been *statistically confirmed* to a conventional standard of significance, given the sample sizes this project’s available queue time permitted. This distinction — between a result that is consistent with a theory and a result that statistically confirms it — is precisely the kind of careful boundary that Chapter 11 formalizes as one of this project’s central, explicitly acknowledged limitations.

Chapter 11

Limitations and Honest Boundaries

This report has, throughout, attempted to state precisely what each result does and does not demonstrate, rather than allowing a favorable-looking number to stand without its necessary qualification. This chapter collects the three most significant such qualifications into one place, so that a reader evaluating this project’s conclusions has a single, explicit reference for the boundaries of what has actually been shown.

11.1 The Gate-Sequence Abstraction: LM05’s Round Trip Is Not a Physical Two-Hop Journey

Section 5.1 flagged this limitation in passing when first introducing LM05’s two-way structure; it is restated here in full, since it bears directly on how the real-hardware results of Chapter 9 should be interpreted.

Every LM05 circuit implemented in this project — across all three levels — represents the protocol’s round trip (Bob’s preparation, Alice’s Control Mode or Message Mode operation, and Bob’s final measurement in Message Mode) as a single sequence of gates applied within *one* circuit object, executed as one continuous job on the simulator or real processor. This is a faithful and correct representation of the protocol’s *logical* structure: the mathematics of Chapter 6 and the measured outcomes of Chapters 7 through 9 accurately reflect what the protocol’s defining equations predict. It is not, however, a representation of a qubit physically leaving a quantum chip, traveling through a real channel to a separate physical location, and returning — the “round trip” exists here as a sequence of operations in time on a single, stationary qubit, not as physical spatial motion between two separated hardware endpoints.

This distinction matters specifically for the interpretation of Chapter 9’s real-hardware results. Those results correctly validate that LM05’s protocol logic behaves as predicted when executed on genuine quantum hardware, subject to that hardware’s real gate and readout noise. They do not validate any claim about what would happen if the two legs of the round trip were separated by a genuine physical channel of some length — for instance, a lossy optical fiber connecting two geographically distant devices — since no such channel-level separation exists in a single-QPU, single-circuit implementation. The channel-loss analysis of Sections 3.8 and 4.8 remains a theoretical treatment only, not one confirmed by this

project’s experimental data, which speaks solely to gate-level and readout-level noise on a fixed, unified quantum processor.

11.2 Finite-Key Sample-Size Asymmetry

This limitation has been demonstrated numerically at every stage of this report, from the binomial-expectation calculations of Section 7.3 through the confidence-interval analysis of Section 9.5, and it is worth restating as a single, unified structural fact. For any given number of raw qubits N , LM05’s Control Mode QBER-testing sample is, in expectation, only $N/4$ — half the size of BB84’s $N/2$ sifted key — because LM05’s security test requires two independent filtering conditions (Alice’s random mode choice, and a subsequent basis match) where BB84’s requires only one.

The practical consequence of this asymmetry, demonstrated concretely in Section 9.5, is that any QBER comparison between the two protocols conducted at matched values of N is not actually comparing statistically equivalent samples: LM05’s QBER estimate at a given N will always carry more sampling uncertainty than BB84’s estimate at the same N , purely as a structural feature of the two protocols, independent of any question about their true underlying error rates. This asymmetry is precisely why Run 2’s apparently contradictory LM05 result (Section 9.4) could be confidently attributed to statistical noise rather than a genuine reversal of the theoretical prediction, and it is why any future attempt to confirm Chapter 5’s theoretical prediction with full statistical confidence would need to scale N specifically with LM05’s smaller effective sample size in mind, rather than simply matching raw qubit counts between the two protocols.

11.3 What This Project Does and Does Not Prove

Bringing together the two limitations above, alongside the statistical findings of Section 10.4, this project’s evidentiary claims can be stated precisely as follows.

This project **does** demonstrate: that both BB84 and LM05 can be correctly implemented as Qiskit circuits, with their logical correctness confirmed by an exact, deterministic zero-QBER result on a noise-free simulator; that a realistic noise model can be successfully injected into these same circuits, producing statistically sensible, calibration-dependent QBER and secure-key-rate behavior; that both protocols can be successfully executed on genuine IBM quantum hardware, with results directionally consistent with the noisy-simulator predictions once the historical nature of the fake-backend calibration snapshots is accounted for; and that the specific numerical outcomes obtained — QBER values, sifted key lengths, and secure key rates — are, in every case examined in this report, explicable by direct appeal to the underlying probability theory and the specific gate-level structure of each protocol’s circuit, rather than being unexplained empirical curiosities.

This project **does not** demonstrate: a statistically significant confirmation, to a conventional threshold such as $p < 0.05$, that LM05’s true QBER genuinely exceeds BB84’s on real hardware — the best available evidence (Section 9.5) is directionally consistent with this claim but falls short of formal significance; any result concerning genuine physical channel

loss, distance-dependent attenuation, or a true multi-location two-way channel for LM05, since no such physical separation exists in this project’s single-QPU circuit implementation; and any claim about LM05’s vulnerability, or lack thereof, to the channel-substitution attack described in Section 4.6, since this project’s scope was limited to protocol-logic validation under gate and readout noise, and did not include an implementation or simulation of an active adversarial party.

Stating these boundaries explicitly is not a weakness of this report but, in keeping with the scientific rigor this report has aimed for throughout, a necessary part of presenting its findings honestly. Chapter 12 now draws the report to its conclusion, summarizing what has been established and outlining the specific, concrete directions in which this work could be extended.

Chapter 12

Conclusion

12.1 Summary of Findings

This report set out to answer a specific question stated at the very beginning: how do two structurally different quantum key distribution protocols, BB84 and LM05, actually behave when taken from theoretical description through to execution on genuine quantum hardware? The preceding eleven chapters have traced a single, continuous line of reasoning toward an answer.

Part I established the physical and mathematical foundations — the qubit as a state vector, the Z and X bases, the Born rule, and the no-cloning theorem — that both protocols depend upon without exception. Part II developed BB84 and LM05 individually, in full mathematical detail, culminating in a direct theoretical comparison that produced a specific, falsifiable prediction: that LM05’s two-way channel structure, and the additional gate its Message Mode encoding requires (established concretely in Section 6.5), should cause it to exhibit a measurably higher quantum bit error rate than BB84 under any realistic source of physical noise.

Part III tested this prediction through three progressively more realistic stages of implementation. The ideal simulator (Chapter 7) confirmed that both protocols’ circuit logic was implemented correctly, exhibiting the deterministic zero-QBER behavior that the underlying mathematics requires in the total absence of noise. The noisy simulator (Chapter 8) confirmed that the noise-injection mechanism itself functioned correctly, and produced a first, statistically inconclusive hint that LM05’s QBER might indeed exceed BB84’s, as predicted. Real IBM quantum hardware (Chapter 9), tested across three independent runs of increasing sample size, produced results directionally consistent with the theoretical prediction at the two largest and most reliable sample sizes, while also surfacing an apparently contradictory result at an intermediate sample size that a rigorous statistical treatment revealed to be an artifact of insufficient data rather than genuine evidence against the theory.

Part IV synthesized these three levels into a single combined analysis (Chapter 10), tracing the precise mathematical progression — from a deterministic model, to a known-parameter stochastic model, to an unknown-parameter stochastic model — that explains why three distinct implementation stages were methodologically necessary, and concluded with a formal two-proportion significance test on the project’s best available data. That

test found the theoretically predicted direction present in the data, but not yet reaching conventional statistical significance. Chapter 11 then stated explicitly the boundaries of what this project’s evidence does and does not support, particularly regarding the gate-sequence-abstraction nature of LM05’s implemented round trip, and the structural sample-size asymmetry between the two protocols’ security-testing subsets.

Taken as a whole, this report’s central finding is best stated with the same precision that has governed every result throughout: the theoretical prediction that LM05 underperforms BB84 under realistic noise is consistently supported, in direction, by every noisy data set gathered in this project, across both simulation and genuine quantum hardware — but it has not yet been confirmed to a conventional standard of statistical significance, a limitation attributable specifically to the finite sample sizes available within this project’s implementation and its access to real quantum hardware compute time, rather than to any flaw in the theoretical prediction or its implementation.

12.2 Suggestions for Future Work

Three concrete directions follow naturally from the limitations identified in Chapter 11, each addressing a specific, identified gap in this project’s current evidence.

Scaling to a larger raw qubit count. The two-proportion z-test of Section 9.5 found a p -value of approximately 0.64 at $N = 500$; achieving conventional statistical significance ($p < 0.05$) for the predicted BB84-versus-LM05 QBER difference would require substantially more data. Given the sample-size asymmetry established in Section 11.2 — LM05’s effective testing sample scaling as $N/4$ against BB84’s $N/2$ — and the pooled error rate observed in Run 3 ($\hat{p} \approx 0.0054$), a rough power calculation suggests that an N on the order of several thousand raw qubits, rather than 500, would likely be needed to shrink the confidence intervals of Section 9.5 enough for them to stop overlapping. This would require either a substantially larger allocation of IBM Quantum compute time than the Open Plan’s free tier provides, or a more efficient batching of circuit submissions within the available queue budget.

Using live, current-calibration noise models rather than historical fake backends. Section 9.6 identified a specific limitation of the FakeManilaV2 and FakeLimaV2 noise models used in Chapter 8: their calibration snapshots are frozen at some point in the past and do not reflect the live, current state of any real device. A natural refinement would be to construct a noise model directly from `ibm_fez`’s own current calibration data (via `AerSimulator.from_backend()` applied to the live backend object, rather than a static fake-backend class), immediately before submitting the corresponding real-hardware job. This would allow Chapter 8’s simulated prediction and Chapter 9’s real-hardware measurement to be compared under matched, contemporaneous calibration conditions, removing the historical-snapshot confound identified in this report.

Investigating measurement-device-independent (MDI) LM05 variants. Section 4.7 noted that the channel-substitution vulnerability described in Section 4.6 is addressed, in the broader literature, by measurement-device-independent protocol variants that remove trust assumptions about measurement devices entirely. This report’s scope, as stated explicitly in Section 11.3, did not extend to implementing or testing any such variant, nor to

simulating an active adversarial party of any kind. A natural extension of this work would implement a simplified MDI-LM05 circuit, and empirically compare its noise sensitivity and key-generation efficiency against the standard LM05 implementation studied throughout this report, providing an empirical basis for evaluating whether the added security guarantee of an MDI variant comes at a measurable performance cost.

Each of these three directions is a specific, well-defined extension of the work presented here, rather than a vague call for "further research" — a distinction consistent with the standard of precision this report has aimed to maintain from its opening chapter to this, its last.

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- [13] Qiskit Aer documentation, <https://qiskit.github.io/qiskit-aer/>, accessed 2026.

Appendix A: Tools and Platforms Used

This appendix documents, in depth, the two categories of software tools and cloud platforms used across the three-level implementation presented in Part III of this report.

A.1 IBM Quantum Platform / Qiskit Runtime

IBM Quantum Platform is IBM’s cloud service for submitting quantum circuits to real, physical superconducting quantum processors. The Python package `qiskit-ibm-runtime` provides the client-side interface used throughout Chapter 9 of this report.

QiskitRuntimeService is the authentication and connection object. Calling `QiskitRuntimeService` with an API token and Cloud Resource Name (CRN) configures all subsequent communication with IBM’s backend infrastructure — listing available devices, checking their queue status, and submitting jobs.

Backends are the named, physical QPUs (for example, `ibm_fez`, used throughout Chapter 9). Each backend object exposes properties including qubit count, native basis gates, and calibration data such as gate error rates, readout error rates, and coherence times.

generate_preset_pass_manager builds the transpilation pipeline that converts an abstract circuit (written using gates such as H , X , Y) into an equivalent circuit expressed only in a backend’s native gate set, respecting the chip’s physical qubit connectivity — the mechanism discussed mathematically in Section 9.2.

SamplerV2 is the modern IBM Runtime primitive used to execute circuits and retrieve measurement-outcome counts, submitting a batch of circuits and returning structured results.

The Open Plan is the free access tier used throughout this project, providing a limited monthly allocation of real QPU time at no cost, shared across a global queue of other free-tier users — the source of the queue waits and occasional job-synchronization errors encountered during Chapter 9’s implementation.

A.2 Qiskit and Qiskit Aer

Qiskit is IBM’s open-source SDK for building, manipulating, and simulating quantum circuits. The `QuantumCircuit` class, used throughout every level of this project’s implemen-

tation, is Qiskit’s core object, representing a sequence of gates and measurements independently of any particular hardware backend.

Qiskit Aer is Qiskit’s high-performance simulation backend, providing `AerSimulator()`, the noise-free simulator used in Chapter 7, and `AerSimulator(noise_model=...)`, the noise-injected simulator used in Chapter 8.

A `NoiseModel`, as used in Chapter 8, encodes — per qubit and per gate — depolarizing error rates for single- and two-qubit gates, readout (measurement) error represented as a confusion matrix, and thermal relaxation parameters (T_1 , T_2) describing coherence decay over time.

FakeManilaV2 and **FakeLimaV2**, imported from `qiskit_ibm_runtime.fake_provider`, are objects that replicate the qubit count, basis gates, and connectivity of the real `ibmq_manila` and `ibmq_lima` devices, using a calibration snapshot frozen at the time of capture — the specific limitation discussed at length in Section 8.2 and revisited in Section 9.6, where it was used to explain the discrepancy between this project’s Level 2 predictions and its Level 3 real-hardware measurements.

Using the same `QuantumCircuit` objects across all three levels — passed to `AerSimulator()` in Chapter 7, to `AerSimulator(noise_model=...)` in Chapter 8, and, after transpilation, to `SamplerV2` for execution on `ibmq_fez` in Chapter 9 — is what allowed this report’s three-level comparison to isolate execution environment as the only variable distinguishing its results, a methodological point noted at several points throughout Part III.

Further Reading

- IBM Quantum Platform: <https://quantum.cloud.ibm.com>
- Qiskit Runtime documentation: <https://quantum.cloud.ibm.com/docs>
- QiskitRuntimeService API reference: <https://quantum.cloud.ibm.com/docs/api/qiskit-ibm-runtime/qiskit-runtime-service>
- Transpilation guide: <https://quantum.cloud.ibm.com/docs/en/guides/transpile>
- Qiskit Aer documentation: <https://qiskit.github.io/qiskit-aer/>
- Fake Provider overview: <https://quantum.cloud.ibm.com/docs/api/qiskit-ibm-runtime/fake-provider>